Zexu Chen

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Research Interests

Computational materials science of structural and functional alloys; phase-field modeling of martensitic transformation and precipitation in NiTi shape memory alloys; GPU-accelerated high-performance computing; machine learning and Bayesian inference for materials property prediction and inverse alloy design.

Education

The Ohio State University, Columbus, OH May 2024 – May 2028 (expected)

Ph.D., Materials Science and Engineering

Advisor: Prof. Yunzhi Wang

The Ohio State University, Columbus, OH Aug 2022 – May 2024

M.S., Materials Science and Engineering

Relevant coursework: Structure and Defects in Materials; Diffusion and Interface Kinetics; Thermodynamics of Materials.

Sichuan University / The Ohio State University Aug 2017 – May 2022

B.Sc., Materials Science and Engineering

Publications

Peer-reviewed journal articles

1. Sriram H., Feng L., **Chen Z.**, Wang Y. (2026). Soft and Hard Confinement Effects on Martensitic Transformation in NiTi Shape Memory Alloys: Insights from Phase-Field Simulations. *Shape Memory and Superelasticity.* https://doi.org/10.1007/s40830-026-00619-3
2. Abe O., Qiu Z., **Chen Z.**, Jinschek J.R., Gouma P.-I. (2021). Effect of Crystallite Size on the Low-Temperature Solid-Solid Phase Transformations in the WO_3 System. *Ceramics International.* https://doi.org/10.1016/j.ceramint.2021.08.254

In preparation

- **Chen Z.**, Wang Y. Regulating mechanical behavior of NiTi shape memory alloys via 1D nanoscale concentration modulations: phase-field study and Gaussian Process Regression surrogate. *Manuscript in preparation, target submission 2026.*

Research Experience

Graduate Researcher, Wang Group, The Ohio State University *Aug 2022 – Present*

Project 1 (lead author): Regulating Mechanical Behavior of NiTi SMA via 1D Nanoscale Concentration Modulations

- Designed and ran 3D phase-field simulations of nanoscale concentration modulations to engineer linear superelastic stress-strain response of NiTi shape memory alloys for biomedical and actuator applications.

- Investigated habit-plane and twinning-plane formation in B19' martensite using martensite crystallography and Khachaturyan microelasticity theory; correlated microstructural descriptors with macroscopic mechanical behavior.
- Developing Gaussian Process Regression surrogate models to map concentration-modulation parameters to stress-strain behavior, enabling accelerated inverse design of NiTi alloy compositions.

Project 2: Taming Martensitic Transformation via Dissolution of Precipitates in NiTi SMA

- Modeled Ni_4Ti_3 precipitation and B2-to-B19' martensitic transformation via multiphase-field simulations using Cahn-Hilliard and time-dependent Ginzburg-Landau evolution equations with explicit nucleation algorithm.
- Quantified effects of dissolution-induced concentration fields on transformation hysteresis and mechanical stability; produced parameter guidelines for collaborating experimentalists.
- Implemented GPU-accelerated phase-field code in C with OpenACC on the Ohio Supercomputer Center; managed large parametric job sweeps via SLURM for composition and load studies.

Earlier project (co-author): Crystallite-Size Effects on Phase Transformations in WO_3

- Synthesized WO_3 nanocrystals via sol-gel method; quantified crystallite size distributions from TEM micrographs to correlate microstructure with low-temperature solid-solid phase transformation behavior.

Team Member, "ML for Superconductivity Prediction", The Ohio State University *Spring 2026*

MATSCEN 7193.02 (Machine Learning for Materials Research), team project (4 members)

- Built XGBoost and GPR pipeline to predict superconducting parameters (λ, T_c) from Eliashberg spectral functions $\alpha^2 F(\omega)$ on the Gibson *et al.* (*npj Comput Mater*, 2026) dataset of 806 DFT-validated superconductors; designed physics-informed features (frequency-weighted $\alpha^2 F/\omega$ bins) aligned with the Eliashberg integral.
- Tuned GPR with Matérn ($\nu = 1.5, 2.5$) + White kernels to capture spectral discontinuities; benchmarked spectral bin resolution (50, 200, 800) against generalization to characterize the curse-of-dimensionality trade-off, identifying physics-informed binning as a more robust route than naive high-resolution sampling.

Teaching Experience

Graduate Teaching Assistant, The Ohio State University *Oct 2023 – Oct 2024*

- Instructed two undergraduate courses: *Introduction to Engineering Materials* and *Modeling and Simulation Lab II*.
- Led weekly recitations, held office hours, and graded assignments and exams for approximately 40 students per course.

Conference Presentations

- **TMS 2026** (Mar 2026). Poster: *Soft and Hard Confinement on NiTi B19'-Insight from Phase Field Modeling.*
- **MS&T 2025** (Oct 2025). Oral: *Regulating Martensitic Transformation in NiTi via Microstructural Engineering.*

- **TMS 2025** (Mar 2025). Poster: *Regulating Stress-Strain Behavior of NiTi SMA via 1D Nanoscale Concentration Modulations.*
- **MS&T 2023** (Oct 2023). Oral: *Fine-tuning Superelastic Behavior of NiTi SMAs via Nanoscale Concentration Modulations.*

Professional Service

Reviewer, *Shape Memory and Superelasticity* (ASM International / Springer) *2026 – Present*

Reviewer, *Metallurgical and Materials Transactions A* (TMS / ASM International / Springer) *2026 – Present*

Technical Skills

Programming: Python (scikit-learn, PyTorch), C, MATLAB, Fortran, Bash
HPC & GPU: OpenACC, CUDA, SLURM, Linux, Git, Ohio Supercomputer Center
Simulation: Phase Field Modeling, Molecular Dynamics, Constitutive Modeling, Kinetic Monte Carlo
Machine Learning: LASSO regression, Bayesian optimization, Gaussian Process Regression, Physics-Informed Neural Networks (PINNs)
Data & Visualization: OVITO, ParaView, PyVista, JupyterLab, MATLAB Advanced Plotting
Domain Expertise: Shape memory alloys, martensitic transformation, ICME, materials informatics
Languages: English (Fluent), Mandarin (Native)

References available upon request.